

SANJIVANI UNIVERSITY



Project Title :

Smart Worker Connection System

Students Name :

1. Uday Shelke(2124UCEM1031)
2. Pranav Pagar(2124UCEM1039)
3. Vedant Kshatriya(2124UCEM1095)
4. Krishna Bhad(2124UCEM1060)
5. Kanchan Jagtap(2124UCEM1095)
6. Atharva Jagtap(2124UCEM1104)
7. Sarthak Harde(2124UCEM1006)

UNDER THE GUIDANCE OF :

Prof: Nitin Chopde Sir

**Department of Computer Science and Engineering,
Sanjivani University,
Rural Education Society, Kopergaon, Maharashtra.**

Problem Statement

Current traffic systems use fixed signals and manual monitoring, which cannot handle real-time traffic, detect violations effectively, or prioritize emergency vehicles. This results in congestion, delays, and reduced road safety. They also lack real-time monitoring and efficient coordination between different traffic junctions, creating a need for an intelligent, automated traffic management system.

Existing Solutions

- **Fixed-Time Traffic Signals** – Signals operate on preset timings without considering real-time traffic conditions.
- **Manual Traffic Control** – Traffic police manage congestion and violations manually.
- **CCTV Surveillance Systems** – Cameras are used for monitoring but require human observation.
- **Basic Sensor-Based Systems** – Use loop detectors or IR sensors to detect vehicle presence, but with limited accuracy and adaptability.
- **E-Challan Systems** – Automatically issue fines for violations, but do not prevent congestion or manage traffic flow.

Objectives

- To develop a system that dynamically controls traffic signals based on real-time traffic density.
- To automatically detect and report traffic violations using AI and camera feeds.
- To enable real-time vehicle tracking across multiple locations
- To provide priority clearance for emergency vehicles like ambulances.
- To create a centralized dashboard for monitoring and managing traffic efficiently.

Proposed Solution

- The proposed system is a Smart AI-Based Traffic Management System that uses Artificial Intelligence and Computer Vision to manage traffic efficiently.
- It dynamically controls traffic signals based on real-time vehicle density, automatically detects traffic violations using cameras, and enables real-time vehicle tracking.
- It also provides priority clearance for emergency vehicles like ambulances by adjusting signals automatically. A centralized dashboard allows authorities to monitor traffic, take quick decisions, and improve road safety.

Technology Stack and Methodology

- **Development Tools**

VS Code – Used to write and run code

- **Detection**

YOLOv8 – Used to detect number Plates

EasyOCR – Used to read text from number plates

OpenCV – Used for camera and image processing

- **Frontend (UI)**

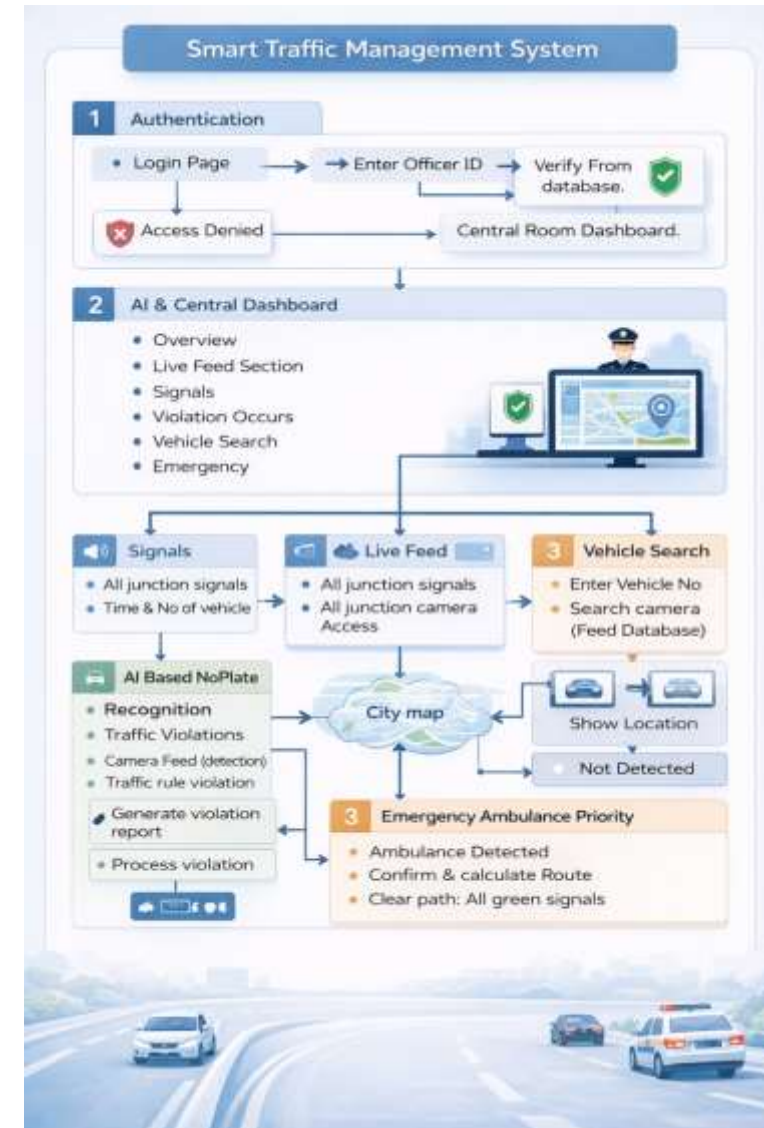
Tkinter – Used to create login page and dashboard

- **Backend**

Python Scripts – Used for detection, processing, and logic

- **Testing**

VS Code Terminal – Used to run and test codeWindows



Impact :-

- Reduces traffic congestion through smart signal control
- Saves time and fuel for commuters
- Improves road safety by detecting violations automatically
- Enables faster movement of emergency vehicles
- Reduces air pollution caused by traffic jams
- Minimizes need for manual traffic control
- Enhances real-time monitoring and decision-making

Future Scope :-

- Integration with smart city infrastructure
- Use of predictive analytics for traffic forecasting
- Mobile app for public traffic updates and alerts
- Integration with GPS and navigation systems
- Expansion to multi-city or nationwide implementation

•Conclusion

The Smart AI-Based Traffic Management System provides an efficient and intelligent solution to modern traffic problems. By using AI and real-time data, it reduces congestion, improves road safety, and ensures faster movement of emergency vehicles. Overall, it makes traffic management smarter, automated, and more reliable, supporting the development of smart cities